

SECTION 1: DOUBLE RIVETED LAP JOINT UNDER TENSION, SHEAR, AND CRUSHING STRESSES

System Parameters: Plate Thickness $t = 15$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 75$ mm, Ultimate Tensile Stress $\sigma_u = 400$ MPa, Ultimate Shear Stress $\tau_u = 320$ MPa, Ultimate Crushing Stress $\sigma_{cu} = 640$ MPa, Factor of Safety FOS = 4.

Design Protocol: Evaluate plate tearing, rivet shearing ($n = 2$), and rivet crushing strengths to determine minimum rupturing force P_u per pitch; compute working load $P = P_u / \text{FOS}$ and evaluate working stresses.

1. Tearing Strength of Plate (P_t)

Plate tearing capacity per pitch length

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_u \\ &= (75 - 25) \times 15 \times 400 \\ &= 300000 \text{ N} \\ \Rightarrow P_t &= 300 \text{ kN} \end{aligned}$$

2. Shearing Strength of Rivets (P_s)

Single shear capacity of 2 rivets per pitch length

$$\begin{aligned} P_s &= n \cdot \frac{\pi}{4}d^2 \cdot \tau_u \\ &= 2 \times \frac{\pi}{4}(25)^2 \times 320 \\ &= 314160 \text{ N} \\ \Rightarrow P_s &= 314.16 \text{ kN} \end{aligned}$$

3. Crushing Strength of Rivets (P_c)

Bearing capacity of 2 rivets per pitch length

$$\begin{aligned}
 P_c &= n \cdot d \cdot t \cdot \sigma_{cu} \\
 &= 2 \times 25 \times 15 \times 640 \\
 &= 480000 \text{ N} \\
 \Rightarrow P_c &= 480 \text{ kN}
 \end{aligned}$$

4. Rupturing Force per Pitch (P_u)

Minimum strength governing joint rupture

$$\begin{aligned}
 P_u &= \min(P_t, P_s, P_c) \\
 &= \min(300, 314.16, 480) \\
 \Rightarrow P_u &= 300 \text{ kN}
 \end{aligned}$$

5. Safe Working Load (P)

Applied force per pitch at FOS = 4

$$\begin{aligned}
 P &= \frac{P_u}{\text{FOS}} \\
 &= \frac{300}{4} \\
 &= 75 \text{ kN} \\
 &= 75000 \text{ N}
 \end{aligned}$$

6. Actual Tensile Stress (σ_t)

Induced tensile stress on main plate

$$\begin{aligned}\sigma_t &= \frac{P}{(p - d)t} \\ &= \frac{75000}{(75 - 25) \times 15} \\ &= 100 \text{ MPa}\end{aligned}$$

7. Actual Shear Stress (τ)

Induced shear stress in rivets

$$\begin{aligned}\tau &= \frac{P}{n \cdot \frac{\pi}{4} d^2} \\ &= \frac{75000}{2 \times \frac{\pi}{4} (25)^2} \\ &= 76.4 \text{ MPa}\end{aligned}$$

8. Actual Crushing Stress (σ_c)

Induced crushing stress on rivets

$$\begin{aligned}\sigma_c &= \frac{P}{n \cdot d \cdot t} \\ &= \frac{75000}{2 \times 25 \times 15} \\ &= 100 \text{ MPa}\end{aligned}$$

Design Output

Rupturing Force $P_u = 300 \text{ kN}$ | Working Load $P = 75 \text{ kN}$ |
Working Stresses: $\sigma_t = 100 \text{ MPa}$, $\tau = 76.4 \text{ MPa}$,
 $\sigma_c = 100 \text{ MPa}$

SECTION 2: EFFICIENCY OF SINGLE AND DOUBLE RIVETED LAP JOINTS

System Parameters: Plate Thickness $t = 6$ mm, Rivet Diameter $d = 20$ mm, Case 1 Pitch $p_1 = 50$ mm ($n = 1$), Case 2 Pitch $p_2 = 65$ mm ($n = 2$), Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 90$ MPa, Crushing $\sigma_c = 180$ MPa.

Design Protocol: Calculate un-punched solid plate strength, tearing, shearing, and crushing strengths for single and double riveted lap joints; compute joint efficiencies.

1. Solid Plate Strength — Single Lap (P)

Strength of un-punched plate ($p = 50$ mm)

$$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 50 \times 6 \times 120 \\ &= 36000 \text{ N} \end{aligned}$$

2. Tearing Strength — Single Lap (P_t)

Net section tensile capacity

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (50 - 20) \times 6 \times 120 \\ &= 21600 \text{ N} \end{aligned}$$

3. Shearing Strength — Single Lap (P_s)

Single shear capacity ($n = 1$)

$$\begin{aligned} P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1 \times \frac{\pi}{4} (20)^2 \times 90 \\ &= 28274 \text{ N} \end{aligned}$$

4. Crushing Strength — Single Lap (P_c)Bearing capacity ($n = 1$)

$$\begin{aligned}
 P_c &= n \cdot d \cdot t \cdot \sigma_c \\
 &= 1 \times 20 \times 6 \times 180 \\
 &= 21600 \text{ N}
 \end{aligned}$$

5. Efficiency — Single Riveted Lap Joint (η_1)

Ratio of minimum strength to solid strength

$$\begin{aligned}
 \eta_1 &= \frac{\min(P_t, P_s, P_c)}{P} \times 100\% \\
 &= \frac{21600}{36000} \times 100\% \\
 &\Rightarrow \eta_1 = 60.0\%
 \end{aligned}$$

6. Solid Plate Strength — Double Lap (P)Strength of un-punched plate ($p = 65 \text{ mm}$)

$$\begin{aligned}
 P &= p \cdot t \cdot \sigma_t \\
 &= 65 \times 6 \times 120 \\
 &= 46800 \text{ N}
 \end{aligned}$$

7. Tearing Strength — Double Lap (P_t)

Net section tensile capacity

$$\begin{aligned}
 P_t &= (p - d)t \cdot \sigma_t \\
 &= (65 - 20) \times 6 \times 120 \\
 &= 32400 \text{ N}
 \end{aligned}$$

8. Shearing Strength — Double Lap (P_s)

Single shear capacity ($n = 2$)

$$\begin{aligned} P_s &= n \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 2 \times \frac{\pi}{4} (20)^2 \times 90 \\ &= 56548 \text{ N} \end{aligned}$$

9. Crushing Strength — Double Lap (P_c)

Bearing capacity ($n = 2$)

$$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_c \\ &= 2 \times 20 \times 6 \times 180 \\ &= 43200 \text{ N} \end{aligned}$$

10. Efficiency — Double Riveted Lap Joint (η_2)

Ratio of minimum strength to solid strength

$$\begin{aligned} \eta_2 &= \frac{\min(P_t, P_s, P_c)}{P} \times 100\% \\ &= \frac{32400}{46800} \times 100\% \\ \Rightarrow \eta_2 &= 69.2\% \end{aligned}$$

Design Output

Single Riveted Lap Efficiency $\eta_1 = 60.0\%$ | Double Riveted Lap Efficiency $\eta_2 = 69.2\%$

SECTION 3: EFFICIENCY OF DOUBLE RIVETED DOUBLE COVER BUTT JOINT

System Parameters: Plate Thickness $t = 20$ mm, Rivet Diameter $d = 25$ mm, Pitch $p = 100$ mm, Allowable Stresses: Tensile $\sigma_t = 120$ MPa, Shear $\tau = 100$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 2.0.

Design Protocol: Evaluate solid plate strength, plate tearing strength, rivet double shear strength ($n = 2$), and rivet crushing strength to determine joint efficiency.

1. Solid Plate Strength (P)

Un-punched solid plate tensile capacity

$$\begin{aligned} P &= p \cdot t \cdot \sigma_t \\ &= 100 \times 20 \times 120 \\ &= 240000 \text{ N} \end{aligned}$$

2. Tearing Strength of Plate (P_t)

Net section tensile strength per pitch

$$\begin{aligned} P_t &= (p - d)t \cdot \sigma_t \\ &= (100 - 25) \times 20 \times 120 \\ &= 180000 \text{ N} \end{aligned}$$

3. Shearing Strength of Rivets (P_s)

Double shear capacity of 2 rivets per pitch ($n = 2$, factor = 2)

$$\begin{aligned} P_s &= 2 \cdot \left(2 \cdot \frac{\pi}{4} d^2 \right) \cdot \tau \\ &= 2 \times \left(2 \times \frac{\pi}{4} (25)^2 \right) \times 100 \\ &= 196350 \text{ N} \end{aligned}$$

4. Crushing Strength of Rivets (P_c)

Bearing capacity of 2 rivets per pitch ($n = 2$)

$$\begin{aligned} P_c &= n \cdot d \cdot t \cdot \sigma_c \\ &= 2 \times 25 \times 20 \times 150 \\ &= 150000 \text{ N} \end{aligned}$$

5. Joint Efficiency (η)

Ratio of minimum strength to un-punched solid plate strength

$$\begin{aligned} \eta &= \frac{\min(P_t, P_s, P_c)}{P} \times 100\% \\ &= \frac{150000}{240000} \times 100\% \\ \Rightarrow \eta &= 62.5\% \end{aligned}$$

Design Output

Double Cover Butt Joint Efficiency $\eta = 62.5\%$ (Governed by Rivet Crushing)

SECTION 4: DOUBLE RIVETED LAP JOINT — ZIG-ZAG RIVETING

System Parameters: Plate Thickness $t = 13 \text{ mm}$, Allowable Stresses: Tensile $\sigma_t = 80 \text{ MPa}$, Shear $\tau = 60 \text{ MPa}$, Crushing $\sigma_c = 120 \text{ MPa}$, Number of Rivets per pitch $n = 2$.

Design Protocol: Determine rivet hole diameter using Unwin’s formula, equate tearing strength to shearing strength to solve for pitch p , and evaluate joint efficiency.

1. Rivet Diameter (d)

Unwin’s formula for $t = 13 \text{ mm} \geq 8 \text{ mm}$

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{13} \\ &= 21.63 \text{ mm} \\ \Rightarrow d &= \mathbf{22 \text{ mm}} \end{aligned}$$

2. Selected Rivet Standard Size

Adopted standard nominal rivet diameter

Adopted Rivet Diameter $d = 22 \text{ mm}$

3. Rivet Pitch (p)

Equating tearing strength P_t to shearing strength P_s

($n = 2$)

$$(p - d)t \cdot \sigma_t = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$

$$(p - 22) \times 13 \times 80 = 2 \times \frac{\pi}{4} (22)^2 \times 60$$

$$1040(p - 22) = 45616$$

$$p - 22 = \frac{45616}{1040}$$

$$p - 22 = 43.86$$

$$p = 65.86 \text{ mm}$$

$$\Rightarrow p = \mathbf{66 \text{ mm}}$$

4. Adopted Pitch Dimension

Selected longitudinal pitch length

Adopted Pitch $p = 66 \text{ mm}$

5. Joint Efficiency (η)

Efficiency based on shear strength vs solid plate strength

$$\eta = \frac{P_s}{p \cdot t \cdot \sigma_t} \times 100\%$$

$$= \frac{45616}{66 \times 13 \times 80} \times 100\%$$

$$= \frac{45616}{68640} \times 100\%$$

$$\Rightarrow \eta = \mathbf{66.5\%}$$

Design Output

Zig-zag Lap Joint: Rivet Diameter $d = 22\text{ mm}$ | Pitch $p = 66\text{ mm}$ | Efficiency $\eta = 66.5\%$

SECTION 5: TRIPLE RIVETED LAP JOINT — ZIG-ZAG PATTERN

System Parameters: Plate Thickness $t = 7$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 120$ MPa, Number of Rivets per pitch $n = 3$.

Design Protocol: For thin plates ($t < 8$ mm), equate shearing strength to crushing strength to determine rivet diameter d ; equate tearing strength to shearing strength to find pitch p ; compute transverse row pitch p_b .

1. Rivet Diameter (d)

Equating shearing strength P_s to crushing strength P_c
($t < 8$ mm)

$$\begin{aligned} n \cdot \frac{\pi}{4} d^2 \cdot \tau &= n \cdot d \cdot t \cdot \sigma_c \\ 3 \times \frac{\pi}{4} d^2 \times 60 &= 3 \times d \times 7 \times 120 \\ 141.37d &= 2520 \\ d &= \frac{2520}{141.37} \\ d &= 17.82 \text{ mm} \\ \Rightarrow d &= \mathbf{19 \text{ mm}} \end{aligned}$$

2. Selected Rivet Standard Size

Adopted standard nominal rivet diameter

Adopted Rivet Diameter $d = 19$ mm

3. Rivet Pitch (p)Equating tearing strength P_t to shearing strength P_s $(n = 3)$

$$(p - d)t \cdot \sigma_t = n \cdot \frac{\pi}{4} d^2 \cdot \tau$$

$$(p - 19) \times 7 \times 90 = 3 \times \frac{\pi}{4} (19)^2 \times 60$$

$$630(p - 19) = 51035$$

$$p - 19 = \frac{51035}{630}$$

$$p - 19 = 81.01$$

$$p = 100.01 \text{ mm}$$

$$\Rightarrow p = 100 \text{ mm}$$

4. Adopted Pitch Dimension

Selected longitudinal pitch length

Adopted Pitch $p = 100 \text{ mm}$ **5. Transverse Row Pitch (p_b)**

Empirical relation for zig-zag riveting

$$p_b = 0.33p + 0.67d$$

$$= 0.33(100) + 0.67(19)$$

$$= 33 + 12.73$$

$$= 45.73 \text{ mm}$$

$$\Rightarrow p_b = 46 \text{ mm}$$

Design Output

Triple Lap Joint: $d = 19\text{ mm}$ | Pitch $p = 100\text{ mm}$ | Row Pitch
 $p_b = 46\text{ mm}$

SECTION 6: SINGLE RIVETED DOUBLE STRAP BUTT JOINT

System Parameters: Main Plate Thickness $t = 10 \text{ mm}$, Allowable Tensile Stress $\sigma_t = 80 \text{ MPa}$, Allowable Shear Stress $\tau = 60 \text{ MPa}$, Double Shear Factor = 1.875.

Design Protocol: Determine rivet diameter d using Unwin’s formula, calculate double shear capacity P_s ($n = 1$), solve for pitch p , compute cover strap thickness t_1 , and evaluate efficiency.

1. Rivet Diameter (d)

Unwin’s formula for $t = 10 \text{ mm}$

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{10} \\ &= 18.97 \text{ mm} \\ \Rightarrow d &= \mathbf{19 \text{ mm}} \end{aligned}$$

2. Double Shear Strength (P_s)

IS code double shear strength for single rivet ($n = 1$)

$$\begin{aligned} P_s &= 1.875 \cdot n \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1.875 \times 1 \times \frac{\pi}{4} (19)^2 \times 60 \\ &= \mathbf{31895 \text{ N}} \end{aligned}$$

3. Rivet Pitch (p)

Equating tearing strength P_t to double shear strength P_s

$$\begin{aligned}(p - d)t \cdot \sigma_t &= P_s \\(p - 19) \times 10 \times 80 &= 31895 \\800(p - 19) &= 31895 \\p - 19 &= \frac{31895}{800} \\p - 19 &= 39.87 \\p &= 58.87 \text{ mm} \\&\Rightarrow p = \mathbf{60 \text{ mm}}\end{aligned}$$

4. Cover Strap Thickness (t_1)

Standard proportion for double cover butt joint

$$\begin{aligned}t_1 &= 0.625t \\&= 0.625 \times 10 \\&= 6.25 \text{ mm} \\&\Rightarrow t_1 = \mathbf{7 \text{ mm}}\end{aligned}$$

5. Joint Efficiency (η)

Ratio of double shear strength to solid plate strength

$$\begin{aligned}\eta &= \frac{P_s}{p \cdot t \cdot \sigma_t} \times 100\% \\&= \frac{31895}{60 \times 10 \times 80} \times 100\% \\&= \frac{31895}{48000} \times 100\% \\&\Rightarrow \eta = \mathbf{66.4\%}\end{aligned}$$

Design Output

Double Strap Butt Joint: $d = 19 \text{ mm}$ | Pitch $p = 60 \text{ mm}$ |
Strap $t_1 = 7 \text{ mm}$ | Efficiency $\eta = 66.4\%$

SECTION 7: BOILER LONGITUDINAL DOUBLE RIVETED BUTT JOINT

System Parameters: Boiler Shell Diameter $D = 1.5 \text{ m} = 1500 \text{ mm}$, Internal Steam Pressure $p_i = 0.95 \text{ N/mm}^2$, Target Longitudinal Efficiency $\eta_l = 75\%$, Allowable Tensile Stress $\sigma_t = 90 \text{ MPa}$, Allowable Crushing Stress $\sigma_c = 140 \text{ MPa}$, Allowable Shear Stress $\tau = 56 \text{ MPa}$, Corrosion Allowance = 1.5 mm.

Design Protocol: Calculate shell thickness t , size rivet diameter d , solve pitch p from double shear strength ($n = 2$), check against maximum pitch p_{max} , and summarize design parameters.

1. Shell Plate Thickness (t)

Thin cylinder formula with corrosion allowance

$$\begin{aligned} t &= \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} + 1.5 \\ &= \frac{0.95 \times 1500}{2 \times 90 \times 0.75} + 1.5 \\ &= \frac{1425}{135} + 1.5 \\ &= 10.55 + 1.5 \\ &= 12.05 \text{ mm} \\ &\Rightarrow t = 13 \text{ mm} \end{aligned}$$

2. Selected Shell Thickness

Adopted standard plate thickness

Adopted Shell Thickness $t = 13 \text{ mm}$

3. Rivet Diameter (d)

Unwin's formula for $t = 13$ mm

$$\begin{aligned}
 d &= 6\sqrt{t} \\
 &= 6\sqrt{13} \\
 &= 21.63 \text{ mm} \\
 \Rightarrow d &= \mathbf{22 \text{ mm}}
 \end{aligned}$$

4. Selected Rivet Diameter

Adopted standard nominal rivet size

Adopted Rivet Diameter $d = 22$ mm

5. Double Shear Strength (P_s)

IS double shear capacity for $n = 2$ rivets per pitch

$$\begin{aligned}
 P_s &= 1.875 \cdot n \cdot \frac{\pi}{4} d^2 \cdot \tau \\
 &= 1.875 \times 2 \times \frac{\pi}{4} (22)^2 \times 56 \\
 &= 79828 \text{ N}
 \end{aligned}$$

6. Rivet Pitch (p)

Calculated pitch from $P_t = P_s$ and maximum pitch check (p_{max})

$$\begin{aligned}(p - d)t \cdot \sigma_t &= P_s \\(p - 22) \times 13 \times 90 &= 79828 \\1170(p - 22) &= 79828 \\p - 22 &= 68.23 \\p &= 90.23 \text{ mm} \\p_{\text{max}} &= 3.5t + 41.28 \\&= 3.5(13) + 41.28 \\&= 86.78 \text{ mm} \\\Rightarrow p &= \mathbf{86 \text{ mm}}\end{aligned}$$

Design Output

Boiler Longitudinal Seam: Shell Thickness $t = 13 \text{ mm}$ | Rivet Diameter $d = 22 \text{ mm}$ | Pitch $p = 86 \text{ mm}$

SECTION 8: PRESSURE VESSEL DOUBLE STRAP BUTT JOINT

System Parameters: Vessel Diameter $D = 1 \text{ m} = 1000 \text{ mm}$, Pressure $p_i = 2.75 \text{ N/mm}^2$, Target Efficiency $\eta_l = 79\%$, Allowable Tensile Stress $\sigma_t = 88 \text{ MPa}$, Allowable Shear Stress $\tau = 64 \text{ MPa}$, Double Shear Factor = 1.8.

Design Protocol: Size shell thickness t , solve rivet diameter d , determine outer and inner row pitches ($p_{\text{outer}}, p_{\text{inner}}$), and verify efficiency.

1. Plate Thickness (t)

Thin cylinder formula

$$\begin{aligned} t &= \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} \\ &= \frac{2.75 \times 1000}{2 \times 88 \times 0.79} \\ &= \frac{2750}{139.04} \\ &= 19.78 \text{ mm} \\ &\Rightarrow t = \mathbf{20 \text{ mm}} \end{aligned}$$

2. Rivet Diameter (d)

Unwin's formula for $t = 20 \text{ mm}$

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{20} \\ &= 26.83 \text{ mm} \\ &\Rightarrow d = \mathbf{28.5 \text{ mm}} \end{aligned}$$

3. Outer Pitch (p) & Efficiency (η)

Selected pitch dimensions and verified efficiency

$p_{outer} = 150 \text{ mm}, \quad p_{inner} = 75 \text{ mm}$

$$\begin{aligned} P_t &= (150 - 28.5) \times 20 \times \dots \\ &= 213840 \text{ N} \\ P &= 150 \times 20 \times 88 \\ &= 264000 \text{ N} \\ \eta &= \frac{213840}{264000} \times 100\% \\ \Rightarrow \eta &= 81\% \end{aligned}$$

Design Output

Pressure Vessel Joint: Thickness t = 20 mm | Rivet Diameter
d = 28.5 mm | Outer Pitch p = 150 mm | Efficiency eta = 81%

SECTION 9: BOILER LONGITUDINAL SEAM DESIGN

System Parameters: Boiler Diameter $D = 1.25 \text{ m} = 1250 \text{ mm}$, Steam Pressure $p_i = 2.5 \text{ N/mm}^2$, Ultimate Tensile Stress $\sigma_u = 420 \text{ MPa}$, Ultimate Crushing Stress $\sigma_{cu} = 650 \text{ MPa}$, Ultimate Shear Stress $\tau_u = 300 \text{ MPa}$, Target Efficiency $\eta_l = 80\%$, Factor of Safety $\text{FOS} = 5$, Corrosion Allowance $= 1.5 \text{ mm}$.

Design Protocol: Calculate working allowable stresses, determine plate thickness t , and compute rivet diameter d .

1. Allowable Working Stresses

Ultimate stress divided by $\text{FOS} = 5$

$$\sigma_t = \frac{\sigma_u}{\text{FOS}}, \quad \sigma_c = \frac{\sigma_{cu}}{\text{FOS}}, \quad \tau = \frac{\tau_u}{\text{FOS}}$$
$$\sigma_t = \frac{420}{5}$$
$$= 84 \text{ MPa}$$
$$\sigma_c = \frac{650}{5}$$
$$= 130 \text{ MPa}$$
$$\tau = \frac{300}{5}$$
$$= 60 \text{ MPa}$$

2. Plate Thickness (*t*)

Thin cylinder formula with corrosion allowance

$$\begin{aligned} t &= \frac{p_i \cdot D}{2\sigma_t \cdot \eta_l} + 1.5 \\ &= \frac{2.5 \times 1250}{2 \times 84 \times 0.80} + 1.5 \\ &= \frac{3125}{134.4} + 1.5 \\ &= 23.25 + 1.5 \\ &= 24.75 \text{ mm} \\ \Rightarrow t &= \mathbf{25 \text{ mm}} \end{aligned}$$

3. Rivet Diameter (*d*)

Unwin’s formula for *t* = 25 mm

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{25} \\ &= 30 \text{ mm} \\ \Rightarrow d &= \mathbf{30 \text{ mm}} \end{aligned}$$

Design Output

Boiler Seam: Allowable Stresses (84, 130, 60 MPa) | Plate Thickness *t* = 25 mm | Rivet Diameter *d* = 30 mm

SECTION 10: BOILER LONGITUDINAL & CIRCUMFERENTIAL JOINTS

System Parameters: Boiler Diameter $D = 1.6 \text{ m} = 1600 \text{ mm}$, Pressure $p_i = 2.5 \text{ N/mm}^2$, Allowable Stresses: Tensile $\sigma_t = 75 \text{ MPa}$, Shear $\tau = 60 \text{ MPa}$, Crushing $\sigma_c = 125 \text{ MPa}$.

Design Protocol: Summarize longitudinal joint plate thickness t and rivet diameter d ; design circumferential joint by calculating total steam thrust F and required rivet count N_c .

1. Longitudinal Joint Parameters

Plate thickness and rivet diameter selection

$$t = 35 \text{ mm}, \quad d = 35.5 \text{ mm}$$

2. Total Steam End Thrust (F)

Axial burst force acting on boiler end cap

$$\begin{aligned} F &= \frac{\pi}{4} D^2 \cdot p_i \\ &= \frac{\pi}{4} (1600)^2 \times 2.5 \\ &= 5.026 \times 10^6 \text{ N} \end{aligned}$$

3. Number of Circumferential Rivets (N_c)

Number of single shear rivets required to resist axial thrust

$$\begin{aligned} N_c &= \frac{F}{\frac{\pi}{4}d^2 \cdot \tau} \\ &= \frac{5.026 \times 10^6}{\frac{\pi}{4}(35.5)^2 \times 60} \\ &= \frac{5.026 \times 10^6}{59390} \\ &= 84.6 \\ \Rightarrow N_c &= 85 \text{ rivets} \end{aligned}$$

Design Output

Longitudinal: t = 35 mm, d = 35.5 mm | Circumferential
Joint Thrust F = 5.026 MN | Rivet Count Nc = 85 rivets

SECTION 11: TIE ROD DOUBLE COVER BUTT JOINT

System Parameters: Tie Rod Cross-Section $b = 200$ mm, $t = 12.5$ mm, Allowable Stresses: Tensile $\sigma_t = 80$ MPa, Shear $\tau = 65$ MPa, Crushing $\sigma_c = 160$ MPa, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d , evaluate maximum tensile load capacity P , compute single rivet double shear strength P_s , and determine required rivet count n .

1. Rivet Diameter (d)

Unwin’s formula for $t = 12.5$ mm

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{12.5} \\ &= 21.21 \text{ mm} \\ \Rightarrow d &= \mathbf{21.5 \text{ mm}} \end{aligned}$$

2. Full Tensile Load (P)

Solid tie-rod tensile capacity

$$\begin{aligned} P &= b \cdot t \cdot \sigma_t \\ &= 200 \times 12.5 \times 80 \\ &= 200000 \text{ N} \\ &= 200 \text{ kN} \end{aligned}$$

3. Single Rivet Double Shear Capacity (P_s)
Shear strength per rivet in double cover butt joint

$$\begin{aligned} P_s &= 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1.875 \times \frac{\pi}{4} (21.5)^2 \times 65 \\ &= 44250 \text{ N} \\ &= 44.25 \text{ kN} \end{aligned}$$

4. Required Rivet Count (n)
Total rivets needed per side of joint

$$\begin{aligned} n &= \frac{P}{P_s} \\ &= \frac{200}{44.25} \\ &= 4.52 \\ \Rightarrow n &= 5 \text{ rivets} \end{aligned}$$

Design Output	Tie Rod Joint: Rivet Diameter $d = 21.5 \text{ mm}$ Tensile Load $P = 200 \text{ kN}$ Required Rivets $n = 5 \text{ rivets}$
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SECTION 12: BRIDGE TIE-BAR LOZENGE (DIAMOND) JOINT

System Parameters: Flat Bar Width $b = 350$ mm, Thickness $t = 20$ mm, Allowable Stresses: Tensile $\sigma_t = 90$ MPa, Shear $\tau = 60$ MPa, Crushing $\sigma_c = 150$ MPa, Double Shear Factor = 1.875.

Design Protocol: Calculate rivet diameter d , evaluate net load capacity P at outer row (1 rivet), determine single rivet double shear strength P_s , compute required number of rivets n , and arrange in diamond pattern.

1. Rivet Diameter (d)

Unwin’s formula for $t = 20$ mm

$$\begin{aligned} d &= 6\sqrt{t} \\ &= 6\sqrt{20} \\ &= 26.83 \text{ mm} \\ \Rightarrow d &= \mathbf{27 \text{ mm}} \end{aligned}$$

2. Net Load Capacity (P)

Tearing strength at critical Section 1-1 (1 rivet hole)

$$\begin{aligned} P &= (b - d)t \cdot \sigma_t \\ &= (350 - 27) \times 20 \times 90 \\ &= 323 \times 1800 \\ &= 581400 \text{ N} \\ &= 581.4 \text{ kN} \end{aligned}$$

3. Single Rivet Double Shear Capacity (P_s)

Shear resistance per rivet

$$\begin{aligned} P_s &= 1.875 \cdot \frac{\pi}{4} d^2 \cdot \tau \\ &= 1.875 \times \frac{\pi}{4} (27)^2 \times 60 \\ &= 64412 \text{ N} \\ &= 64.4 \text{ kN} \end{aligned}$$

4. Total Rivet Count (n)

Total rivets required for lozenge arrangement

$$\begin{aligned} n &= \frac{P}{P_s} \\ &= \frac{581.4}{64.4} \\ &= 9.02 \\ \Rightarrow n &= 9 \text{ rivets} \end{aligned}$$

5. Lozenge Rivet Layout

Symmetric diamond pattern from outer row to inner row

Arranged in Diamond Pattern: Row 1 (1 rivet), Row 2 (2 rivets), Row 3 (3 rivets), Row 4 (3 rivets)

Design Output

Bridge Lozenge Joint: $d = 27 \text{ mm}$ | Net Capacity $P = 581.4 \text{ kN}$ | Total Rivets $n = 9$ (Pattern: 1, 2, 3, 3)

SECTION 13: FLAT TIE-BAR LAP JOINT (DIAMOND PATTERN)

System Parameters: Flat Bar Width $b = 200$ mm, Thickness $t = 10$ mm, Rivet Shank Diameter $d = 24$ mm, Rivet Hole Diameter $d_h = 25.5$ mm, Allowable Stresses: Tensile $\sigma_t = 112$ MPa, Shear $\tau = 84$ MPa, Crushing $\sigma_c = 200$ MPa.

Design Protocol: Arrange 4 rivets in diamond pattern (1, 1, 2); evaluate net tearing strength at outer single-rivet row, and compute joint efficiency.

1. Rivet Layout & Pattern

Specified rivet arrangement on flat tie-bar

n = 4 rivets arranged in Diamond Pattern (Row 1: 1, Row 2: 1, Row 3: 2)

2. Net Tearing Strength (P_t)

Tearing capacity at Section 1-1 with hole diameter $d_h = 25.5$ mm

$$\begin{aligned} P_t &= (b - d_h)t \cdot \sigma_t \\ &= (200 - 25.5) \times 10 \times 112 \\ &= 174.5 \times 1120 \\ &= 195440 \text{ N} \\ &= 195.44 \text{ kN} \end{aligned}$$

3. Solid Strength (*P*) & Efficiency (*η*)
Efficiency based on net tearing capacity

$$P = b \cdot t \cdot \sigma_t, \quad \eta = \frac{P_t}{P} \times 100\%$$
$$P = 200 \times 10 \times 112$$
$$= 224000 \text{ N}$$
$$\eta = \frac{195440}{224000} \times 100\%$$
$$\Rightarrow \eta = 87.3\%$$

Design Output

Flat Lap Joint (Diamond Pattern): Net Tearing Strength *P*_t =
195.44 kN | Joint Efficiency *η* = 87.3%

SECTION 14: ECCENTRIC BRACKET WITH VERTICAL RIVET LINE

System Parameters: Plate Thickness $t = 25$ mm, Eccentric Load $P = 50$ kN = 50000 N, Eccentricity $e = 400$ mm, Rivet Spacing $C = 100$ mm, Number of Rivets $n = 4$ in vertical line, Allowable Shear Stress $\tau = 65$ MPa, Allowable Crushing Stress $\sigma_c = 120$ MPa.

Design Protocol: Calculate direct shear force F_{s1} per rivet; compute secondary shear force F_{s2} due to moment Pe ; find max resultant shear force F_R , and solve for rivet diameter d .

1. Direct Shear Force (F_{s1})

Uniform load distribution across 4 rivets

$$\begin{aligned}
 F_{s1} &= \frac{P}{n} \\
 &= \frac{50000}{4} \\
 &= 12500 \text{ N} \\
 &= 12.5 \text{ kN}
 \end{aligned}$$

2. Secondary Shear Force (F_{s2})

Torsional shear force on outermost rivet ($r_{\max} = 150 \text{ mm}$)

$$F_{s2} = \frac{P \cdot e \cdot r_{\max}}{\sum r_i^2} r_i^2$$

$$\sum r_i^2 = 2(150^2 + 50^2)$$

$$= 2(22500 + 2500)$$

$$= 50000 \text{ mm}^2$$

$$F_{s2} = \frac{50000 \times 400 \times 150}{50000}$$

$$= 60000 \text{ N}$$

$$= 60 \text{ kN}$$

3. Maximum Resultant Shear Force (F_R)

Direct plus secondary shear force (co-linear theta = 0 deg)

$$F_R = F_{s1} + F_{s2}$$

$$= 12.5 + 60$$

$$= 72.5 \text{ kN}$$

$$= 72500 \text{ N}$$

4. Rivet Diameter (d)

Equating F_R to rivet single shear strength

$$F_R = \frac{\pi}{4} d^2 \cdot \tau$$

$$72500 = \frac{\pi}{4} d^2 (65)$$

$$d^2 = \frac{4 \times 72500}{\pi \times 65}$$

$$= \frac{290000}{204.2}$$

$$= 1420.2$$

$$d = 37.68 \text{ mm}$$

$$\Rightarrow d = 38 \text{ mm}$$

Design Output

Eccentric Bracket: Direct Shear $F_{s1} = 12.5 \text{ kN}$ | Secondary Shear $F_{s2} = 60 \text{ kN}$ | Resultant $F_R = 72.5 \text{ kN}$ | $d = 38 \text{ mm}$

SECTION 15: ECCENTRICALLY LOADED BRACKET JOINT

System Parameters: Eccentric Load $P = 45 \text{ kN}$, Allowable Shear Stress $\tau \leq 40 \text{ MPa}$.

Design Protocol: Evaluate direct and secondary shear forces to determine maximum resultant shear load $F_R = 15.2 \text{ kN}$; size rivet diameter d .

1. Maximum Resultant Shear Force (F_R)

Critical rivet resultant load from direct and secondary shear

$$F_R = 15.2 \text{ kN} = 15200 \text{ N}$$

2. Rivet Diameter (d)

Single shear area calculation for allowable stress $\tau = 40 \text{ MPa}$

$$\begin{aligned} d &= \sqrt{\frac{4F_R}{\pi\tau}} \\ &= \sqrt{\frac{4 \times 15200}{\pi \times 40}} \\ &= \sqrt{\frac{60800}{125.66}} \\ &= \sqrt{483.8} \\ &= 22.00 \text{ mm} \\ &\Rightarrow d = \mathbf{22 \text{ mm}} \end{aligned}$$

Design Output

Eccentric Bracket (45 kN Load): Critical Resultant FR =
15.2 kN | Rivet Diameter d = 22 mm

SECTION 16: MAXIMUM SAFE ECCENTRIC LOAD CAPACITY

System Parameters: Number of Rivets $n = 4$, Rivet Diameter $d = 20 \text{ mm}$, Working Shear Stress $\tau = 100 \text{ MPa}$.

Design Protocol: Determine maximum single rivet shear load capacity $F_{R,\max}$; express critical resultant force as a function of external load P ($F_R = 0.966P$); solve maximum safe load P .

1. Single Rivet Shear Capacity ($F_{R,\max}$)
Maximum permissible shear force per rivet

$$\begin{aligned} F_{R,\max} &= \frac{\pi}{4} d^2 \cdot \tau \\ &= \frac{\pi}{4} (20)^2 \times 100 \\ &= 31416 \text{ N} \\ &= 31.416 \text{ kN} \end{aligned}$$

2. Resultant Load Relation
Critical rivet resultant load expressed in terms of load P

$$F_R = 0.966P$$

3. Maximum Safe Load (*P*)

Equating *F_R* to single rivet shear capacity

$$P = \frac{F_{R,max}}{0.966}$$
$$= \frac{31.416}{0.966}$$
$$= 32.52 \text{ kN}$$
$$\Rightarrow P = 32.5 \text{ kN}$$

Design Output

Eccentric Load Capacity: Single Rivet Capacity *F_{R,max}* = 31.416 kN | Maximum Safe Load *P* = 32.5 kN

SECTION 17: ECCENTRIC COLUMN BRACKET JOINT

System Parameters: Number of Rivets $n = 6$, Eccentric Load $P = 60 \text{ kN} = 60000 \text{ N}$, Eccentricity $e = 200 \text{ mm}$, Allowable Shear Stress $\tau \leq 150 \text{ MPa}$.

Design Protocol: Calculate primary direct shear F_{s1} and secondary shear F_{s2} ; vectorially combine to get critical resultant shear force F_R ; solve rivet diameter d .

1. Primary & Secondary Shear Forces

Direct load per rivet and torsional secondary load

$$F_{s1} = \frac{P}{n}, \quad F_{s2} = \frac{P \cdot e \cdot r_{\max}}{\sum r_i^2} r_i^2$$
$$F_{s1} = \frac{60}{6}$$
$$= 10 \text{ kN}$$
$$F_{s2} = 41.5 \text{ kN}$$

2. Maximum Resultant Shear Force (F_R)

Vector addition of primary and secondary shear

$$F_R = 51.5 \text{ kN} = 51500 \text{ N}$$

3. Rivet Diameter (d)Equating F_R to single shear capacity

$$\begin{aligned}
 d &= \sqrt{\frac{4F_R}{\pi\tau}} \\
 &= \sqrt{\frac{4 \times 51500}{\pi \times 150}} \\
 &= \sqrt{\frac{206000}{471.24}} \\
 &= \sqrt{437.1} \\
 &= 20.91 \text{ mm} \\
 &\Rightarrow d = 21 \text{ mm}
 \end{aligned}$$

Design Output

6-Rivet Column Bracket: $F_{s1} = 10 \text{ kN}$ | $F_{s2} = 41.5 \text{ kN}$ |
Resultant $F_R = 51.5 \text{ kN}$ | Rivet Diameter $d = 21 \text{ mm}$

SECTION 18: ECCENTRICALLY LOADED INCLINED BRACKET JOINT

System Parameters: Number of Rivets $n = 4$ (A, B, C, D , vertical line, pitch 60 mm), Inclined Load $P = 100 \text{ kN} = 100000 \text{ N}$ at 30^{deg} to horizontal, Eccentricity $e = 150 \text{ mm}$, Allowable Shear Stress $\tau = 80 \text{ MPa}$, Allowable Bending Stress $\sigma_b = 125 \text{ MPa}$, Plate Width $b = 240 \text{ mm}$.

Design Protocol: Resolve load into vertical and horizontal components (P_V, P_H); calculate max resultant shear force F_R on critical rivet D to size diameter d ; determine bending moment M on bracket plate to calculate plate thickness t .

1. Vertical & Horizontal Load Components

Resolving inclined 100 kN load at 30 degrees

$$P_V = P \sin(30^{\text{deg}}), \quad P_H = P \cos(30^{\text{deg}})$$
$$P_V = 100 \sin(30^{\text{deg}})$$
$$= 50 \text{ kN}$$
$$P_H = 100 \cos(30^{\text{deg}})$$
$$= 86.6 \text{ kN}$$

2. Critical Resultant Shear (F_R) & Rivet Diameter (d)

Resultant force on rivet D and single shear sizing

$$F_R = 42.4 \text{ kN} = 42400 \text{ N}$$

$$\begin{aligned} d &= \sqrt{\frac{4F_R}{\pi\tau}} \\ &= \sqrt{\frac{4 \times 42400}{\pi \times 80}} \\ &= \sqrt{\frac{169600}{251.33}} \\ &= \sqrt{674.8} \\ &= 25.98 \text{ mm} \\ \Rightarrow d &= \mathbf{26 \text{ mm}} \end{aligned}$$

3. Bending Moment (*M*) & Bracket Thickness (*t*)

Bending stress analysis of bracket plate

$$M = P_H \cdot e, \quad \sigma_b = \frac{6M}{t \cdot b^2}$$
$$M = 86.6 \times 10^3 \times 150$$
$$= 1.30 \times 10^7 \text{ N} \cdot \text{mm}$$
$$125 = \frac{6 \times 1.30 \times 10^7}{t \times (240)^2}$$
$$= \frac{7.80 \times 10^7}{57600t}$$
$$= \frac{1354.17}{t}$$
$$t = \frac{1354.17}{125}$$
$$= 10.83 \text{ mm}$$
$$\Rightarrow t = 14 \text{ mm}$$

Design Output

Inclined Bracket Joint: PV = 50 kN, PH = 86.6 kN | Critical
FR = 42.4 kN | d = 26 mm | Plate Thickness t = 14 mm